



INTERNAL

Enterprise Context Engine Guide

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Enterprise Context Engine Guide

What it is

The **Enterprise Context Engine** (`app/Services/Platform/Context/EnterpriseContextEngine.php`) builds **one normalized AI context object** from every platform source so that every AI adapter and the QynVA operator consume the *same* context. There is exactly one place that assembles enterprise context — no duplicated context-building anywhere.

It is a pure **read-model**: it reuses existing deterministic services and writes nothing.

What it assembles

Section	Source (reused)
<code>workspace</code>	the resolved <code>Project</code> (uuid, name)
<code>user + permissions</code>	<code>AiWorkspaceContextResolver::effectivePermissions()</code>
<code>cross_module</code>	<code>CrossModuleOrchestrator::gather()</code> (monitoring, assets, automation, knowledge, incidents, notifications, compliance — via the AI Adapter Registry, no module branching)
<code>timeline</code>	<code>GlobalTimelineService::build()</code>
<code>graph</code>	<code>KnowledgeGraphService::build()</code> (Knowledge Graph v2)
<code>search</code>	<code>EnterpriseSearchService::search()</code> (only when a <code>query</code> option is supplied)
<code>summary</code>	deterministic digest (counts only — no fabrication)

The Monitoring/Assets/Automation/Knowledge/Incidents/Notifications/Compliance dimensions arrive through the cross-module gather, which iterates the AI Adapter Registry and respects workspace entitlements.

Usage

```
$context = app(EnterpriseContextEngine::class)→build($project, $user, [
    'query'      ⇒ 'high cpu on web tier',    // optional → includes enterprise search
    'exclude'    ⇒ ['qynva'],                // optional → skip a module in the gather
    'include'    ⇒ ['workspace','cross_module','timeline','graph','search'], // optional su
    'timeline_limit' ⇒ 25,
]);
```

The returned object is bounded for predictable size and is consumed by:

- **QynVA** (the Enterprise AI Operator) — reasons over the whole context.

- **Executive Intelligence** — the executive AI summary narrates a deterministic dashboard built from the same underlying read-models.
- Any future AI adapter — call the engine instead of hand-assembling context.

Design rules

- **Reuse, don't duplicate** — the engine never re-queries data a module service already exposes.
- **Recursion-safe** — QynVA excludes itself from the cross-module gather; the QynVA adapter's own `buildContext()` is registry-introspection only (no gather), so context building cannot recurse.
- **Honest** — counts and real rows only; nothing is fabricated. The `summary` block is computed from the assembled sections.
- **Workspace-scoped + RBAC** — permissions are part of the context so downstream AI can respect them.